

# Spatial Epidemiology of Heat-Related Emergency Department Visits in an Aging Urban Population: A Retrospective Cohort and Predictive Surveillance Model

*Dissertation — Doctor of Public Health*

A. Candidate · Epidemiology · 2026

*SAMPLE EXEMPLAR — FOR PRACTICE AND STRUCTURE ONLY. This is a fictional, abbreviated example created by the Emerging Technologies Laboratory to demonstrate how a thesis or dissertation is organized and to give you something to defend in The Prep Room. Your institution and program almost certainly have their own required template, formatting, and submission rules. Always follow your university's official guidelines over this example. Citations shown here are illustrative of format only and are not real sources.*

## Abstract

Extreme heat is a growing driver of preventable morbidity, and older adults in dense urban settings are disproportionately affected. This dissertation analyzes five years of emergency department records ( $n = 48,210$  heat-related visits) linked to neighborhood-level temperature, social vulnerability, and green-space data to identify spatial risk patterns and to build a predictive surveillance model. Heat-related visits clustered in low-income, low-canopy census tracts, and a gradient-boosted model using forecast temperature and tract-level vulnerability predicted next-day surge with useful accuracy. Findings support targeted, neighborhood-level heat preparedness over uniform citywide alerts.

## Chapter 1: Introduction

Climate-driven heat events are increasing in frequency and severity, and their health burden falls unevenly across a city. Older adults, people with chronic conditions, and residents of neighborhoods with little tree canopy or air conditioning bear the greatest risk, yet most heat warning systems treat a city as a single uniform unit.

This dissertation asks where heat-related emergency visits concentrate, which neighborhood factors explain that concentration, and whether a tract-level model can forecast daily surges accurately enough to guide preparedness. The goal is to move from broad alerts to targeted action.

## Chapter 2: Literature Review

Epidemiologic studies consistently link ambient heat to cardiovascular and renal emergencies, and social vulnerability indices are associated with worse outcomes. However, much prior work analyzes effects at the city or county level, masking the within-city inequities that determine where resources should go.

The reviewed literature also rarely couples descriptive spatial analysis with an operational prediction tool. This study addresses both gaps by pairing cluster detection with a deployable surveillance model.

## Chapter 3: Methods

We assembled a retrospective cohort of heat-season emergency department visits across one metropolitan area over five years, geocoded to census tract and linked to daily gridded temperature, a social vulnerability index, and satellite-derived canopy cover. Spatial clustering was assessed with local indicators of spatial association.

A gradient-boosted classifier was trained to predict next-day elevated visit counts at the tract level using forecast temperature, vulnerability, canopy, and lagged visit history. Performance was evaluated with time-based cross-validation to avoid leakage, reporting area under the curve and calibration.

## Chapter 4: Results

Heat-related visits were significantly clustered, with persistent high-risk clusters in low-income, low-canopy tracts. Social vulnerability and canopy cover remained associated with visit rates after adjusting for temperature.

The predictive model achieved strong discrimination on held-out time periods and was well calibrated, with the largest gains in the highest-vulnerability tracts where baseline rates were elevated.

## **Chapter 5: Discussion**

Results indicate heat risk is spatially structured and predictable, supporting neighborhood-targeted interventions such as cooling-center placement and outreach in identified clusters rather than uniform citywide messaging.

Limitations include reliance on emergency department visits as a proxy for heat morbidity, possible misclassification of heat-related diagnoses, and single-city generalizability. Ethical use of predictive surveillance, including avoiding stigmatization of flagged neighborhoods, is discussed.

## **Chapter 6: Conclusion**

This dissertation mapped the spatial inequity of heat-related emergencies and demonstrated a calibrated, tract-level forecasting tool. The contribution is a practical bridge from descriptive epidemiology to actionable preparedness. Future work should validate across multiple cities and integrate the model into a live public health alerting workflow.

## **References**

- Author, A. B. (Year). Heat and cardiovascular emergencies. *Environmental Health Perspectives*, Vol(Issue), pages.
- Author, C. D. (Year). Social vulnerability and heat mortality. *American Journal of Public Health*, Vol(Issue), pages.
- Author, E. F. (Year). Urban canopy and temperature. *Landscape and Urban Planning*, Vol(Issue), pages.
- Author, G. H. (Year). Spatial cluster detection methods. *Spatial Statistics*, Vol(Issue), pages.
- Author, I. J. (Year). Machine learning for syndromic surveillance. *JAMIA*, Vol(Issue), pages.